EXPERIMENT:9 FILE HANDLING IN C

Q1. Write a program to create a new file and write text into it.

Algorithm:

1* Start

2* Declare a file pointer FILE *fp.

3* Open a file in write mode using: fp = fopen("output.txt", "w");

4* If file opening fails (fp == NULL), show an error message and exit.

5* Ask the user to enter text.

6* Read the text from the user.

7* Write the text into the file using fprintf() or fputs().

8* Close the file using fclose(fp).

9*bDisplay a message that the file has been written successfully.

10* Stop

Code:

```
#include <stdio.h>

int main() {
    FILE *fp;
    char text[200];

    // Open file in write mode
    fp = fopen("output.txt", "w");

    // Check if file opened successfully
    if (fp == NULL) {
        printf("Error! Could not create file.\n");
        return 1;
    }

    // Get text from user
    printf("Enter text to write into the file:\n");
    fgets(text, sizeof(text), stdin);

    // Write text into file
    fprintf(fp, "%s", text);

    // Close file
    fclose(fp);
```

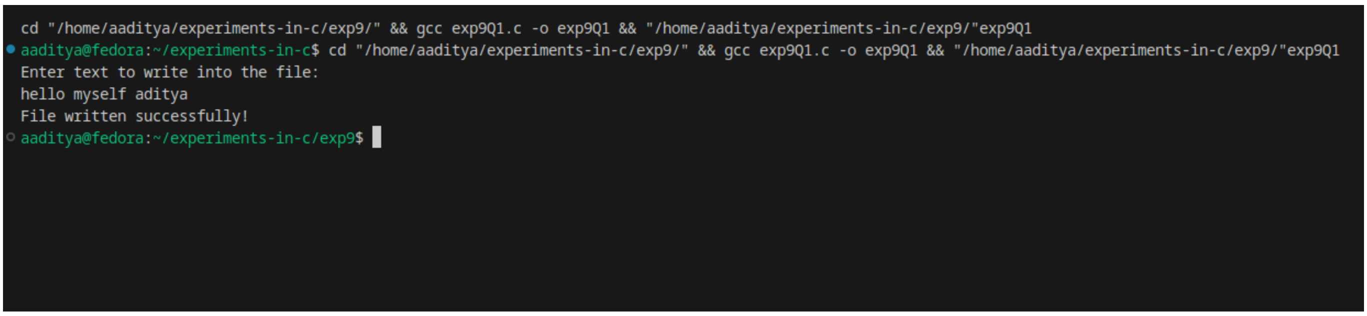
```
    printf("File written successfully!\n");

    return 0;
}
```

Flowchart:

```
      ┌──────────────┐
      │    Start     │
      └──────┬───────┘
             │
             ▼
    ┌──────────────────┐
    │ Declare FILE *fp │
    └────────┬─────────┘
             │
             ▼
```

┌──────────────────┐ │ fp = fopen("output.txt","w") │ └────────┬─────────┘ │ ┌──────▼──────┐ │ fp == NULL ? │ └──────┬──────┘ │Yes ▼ ┌──────────────┐ │ Print error & Stop │ └──────────────┘ │No ▼ ┌──────────────┐ │ Ask user to enter text │ └──────┬───────┘ │ ▼ ┌──────────────┐ │ Write text to the file │ └──────┬───────┘ │ ▼ ┌──────────────┐ │ Close file using fclose │ └──────┬───────┘ │ ▼ ┌──────────────┐ │ Print success message│ └──────┬───────┘ │ ▼ ┌──────────┐ │ Stop │ └──────────┘

Output:



Q2.Open an existing file and read its content character by character and then close the file.

Flowchart:

1-Start

2-Declare a file pointer FILE *fp.

3-Open the file in read mode using: fp = fopen("input.txt", "r");

4-If the file fails to open (fp == NULL), display an error message and stop.

5-Declare a character variable ch.

6-Read a character from the file using: ch = fgetc(fp)

7-While ch != EOF

8-Display the character on the screen

9-Read the next character

10-After reading all characters, close the file using fclose(fp).

11-Stop

Code:

```
#include <stdio.h>

int main() {
    FILE *fp;
    char ch;

    // Open the file in read mode
    fp = fopen("input.txt", "r");

    // Check if file opened successfully
    if (fp == NULL) {
        printf("Error! Could not open file.\n");
        return 1;
    }

    // Read and display characters until EOF
    while ((ch = fgetc(fp)) != EOF) {
        putchar(ch);
    }

    // Close the file
    fclose(fp);

    return 0;
}
```

Flowchart:

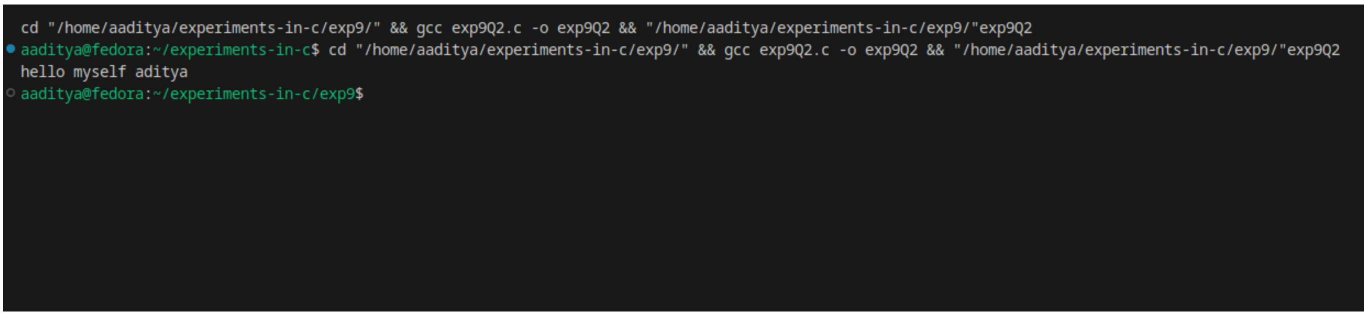
```
        ┌──────────────┐
        │    Start     │
```

```
    └──────────────┘
           │
           │
           ▼
  ┌─────────────────────┐
  │ Declare FILE *fp    │
  └─────────────────────┘
         │
         │
         ▼
```

┌──────────────────────────────────┐ │ fp = fopen("input.txt","r") │
└─────────────┬────────────────────┘ │ ┌────────▼─────────┐ │ fp == NULL ? │
└─────────┬───────────┘ │Yes ▼ ┌──────────────────────┐ │ Print error & Stop │
└────────────────────────┘ │No ▼ ┌──────────────────────┐ │ Read ch = fgetc │
└──────────┬───────────┘ │ ┌────────▼──────────┐ │ ch == EOF ? │
└─────────┬──────────────┘ │Yes ▼ ┌──────────────────────┐ │ Close file fclose │
└──────────────┬───────────┘ │ ▼ ┌─────────────────┐ │ Stop │ └─────────────────┘

```
          │No
          ▼
 ┌─────────────────────┐
 │ Print character ch  │
 └──────────┬──────────┘
            │
            └──▶ Go back to
                 Read next ch
```

Output:


```
cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q2.c -o exp9Q2 && "/home/aaditya/experiments-in-c/exp9/"exp9Q2
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q2.c -o exp9Q2 && "/home/aaditya/experiments-in-c/exp9/"exp9Q2
hello myself aditya
aaditya@fedora:~/experiments-in-c/exp9$
```

Q3.Open a file read its content line by line and display each line on the console:

Algorithm:

1-Start

2-Declare a file pointer FILE *fp.

3-Declare a string buffer (e.g., char line[200]😉.

4-Open the file in read mode using: fp = fopen("input.txt", "r");

5-If fp == NULL, print an error message and stop.

6-Read a line using: fgets(line, sizeof(line), fp)

7-While the line is not NULL:

8-Display the line on the console using printf("%s", line);

9-Read the next line using fgets()

10-Close the file using fclose(fp).

11-Stop

Code:

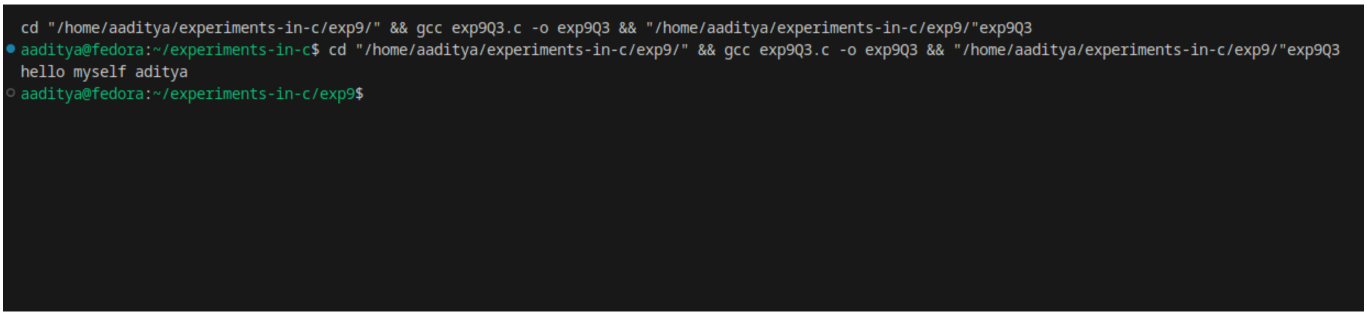
```
#include <stdio.h>

int main() {
    FILE *fp;
    char line[200];

    // Open file in read mode
    fp = fopen("input.txt", "r");

    if (fp == NULL) {
        printf("Error! Could not open file.\n");
        return 1;
    }

    // Read and display each line
    while (fgets(line, sizeof(line), fp) != NULL) {
        printf("%s", line);
    }

    // Close the file
    fclose(fp);

    return 0;
}
```

Flowchart:

```
              ┌──────────────┐
              │    Start     │
              └──────┬───────┘
                     │
                     ▼
        ┌────────────────────────┐
        │ Declare FILE *fp, line │
```

```
        └─────────────────────────┘
                     │
                     │
                     ▼
   ┌───────────────────────────────────┐
   │ fp = fopen("input.txt", "r")      │
   └───────────────────────────────────┘
                    │
                    │
           ┌────────▼────────┐
           │ fp == NULL ?    │ │
           └─────────────────┘
                    │
                    │Yes
                    ▼
        ┌──────────────────────────┐
        │ Print error & Stop       │
        └──────────────────────────┘
                    │No
                    ▼
     ┌───────────────────────────────────┐
     │ Read line = fgets(line, size,fp) │
     └───────────────────────────────────┘
                 │
                 │
        ┌────────▼───────────┐
        │ line == NULL ?     │ │
        └────────────────────┘
                 │
                 │Yes
                 ▼
        ┌─────────────────────────┐
        │ Close file (fclose)     │
        └─────────────────────────┘
                   │
                   │
                   ▼
               ┌──────────────┐
               │     Stop     │ │
               └──────────────┘

                 │No
                 ▼
       ┌──────────────────────────────┐
       │ Print the line (printf)      │
       └──────────────────────────────┘
                  │
                  │
                  └───► Go back to read next line
```

Output:


cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q3.c -o exp9Q3 && "/home/aaditya/experiments-in-c/exp9/"exp9Q3
aaditya@fedora:~/experiments-in-c$ cd "/home/aaditya/experiments-in-c/exp9/" && gcc exp9Q3.c -o exp9Q3 && "/home/aaditya/experiments-in-c/exp9/"exp9Q3
hello myself aditya
aaditya@fedora:~/experiments-in-c/exp9$
